

Lecture Notes 7

The Short Run (part 2)

Intermediate Macroeconomics, EC2211

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Contents and literature

- Propagation of economic fluctuations
- Multipliers
- The AS-AD model
- The Taylor rule
- The time inconsistency problem of monetary policy

Literature:

- Jones (2024), chapters 9-13
- Giannone and Primiceri (2024), "The drivers of post-pandemic inflation"
- Nakamura, Riblier and Steinsson (2025), "Beyond the Taylor Rule"

Propagation of economic fluctuations

Propagation of economic fluctuations

Do economic fluctuations reflect important failures in the economic system?

- Can fluctuations be avoided?
- What policies are most effective to stabilize the economy?

Some contrasting views:

- Keynesian/standard: market failures, "animal spirits" – fluctuations are costly
- Real business cycle theory: fluctuations are the economy's optimal response to changes in technology
- Schumpeter: "creative destruction", fluctuations may be necessary to promote entry and innovation

Propagation of economic fluctuations

Strong **co-movements** between economic variables

In a typical recession:

- output and income fall
- consumption falls (or grows slowly), but less than income
- investment falls sharply
- employment falls and unemployment increases
- imports fall

TABLE 10.1

Changes in Key Macroeconomic Variables: Previous Recessions, the Great Recession, and the Covid-19 Recession

	Average of previous recessions since 1950	Great Recession	Covid-19 Recession
GDP	−1.7%	−4.7%	−9.6%
Nonfarm employment	−2.5%	−6.3%	−11.9%
Unemployment rate	2.5 pp	4.5 pp	9.4 pp
<i>Components of GDP</i>			
Consumption	0.4%	−3.4%	−10.6%
Investment	−14.4%	−34.0%	−16.5%
Government purchases	1.2%	5.5%	2.6%
Exports	−1.5%	−10.3%	−24.1%
Imports	−4.2%	−18.7%	−20.2%

Note: pp = percentage points.
Source: The FRED database.

Propagations of economic fluctuations

In a recession:

- income is low
- households may become cautious and pessimistic and reduce consumption
- businesses may reduce investment and hiring
- all this may reduce output and income further

Propagation of economic fluctuations

Consider the IS curve that we derived in the previous lecture notes (but ignore imports and exports for now):

$$Y_t = C_t + G_t + I_t$$

$$Y_t = \underbrace{\alpha_C \bar{Y}_t}_{C_t} + \underbrace{\alpha_G \bar{Y}_t}_{G_t} + \underbrace{[\alpha_I - \beta(r_t - \bar{r})] \bar{Y}_t}_{I_t}$$

Suppose, for example, that there is a temporary increase in α_G so that government expenditure rises from G_t to G'_t but that the interest rate is held fixed

How is the economy affected?

Propagation of economic fluctuations

If α_G increases so that government expenditure rises by ΔG , all else equal:

- Output also increases by ΔG , i.e. $\Delta Y = \Delta G$
- Private consumption and investment are not affected, i.e. $\Delta C = \Delta I = 0$

In the the real world, we typically see co-movements between the variables over the business cycle

- Does higher G induce higher Y which in turn leads to higher C ?
- Evidence on this is mixed, but let us allow for this in the model
- And let us also allow for taxes and transfers

Propagation of economic fluctuations

In the basic version of the IS curve, there was no *propagation* of demand shocks

A typical Keynesian analysis would assume that consumption depends on disposable income:

$$C_t = \alpha_C \bar{Y}_t + \xi[(Y_t - T_t) - (\bar{Y}_t - \bar{T}_t)]$$

Less complicated than it looks!

- First term on right-hand side is the same as before
- ξ is a parameter indicating how sensitive consumption is to disposable income
- T_t denotes households' tax payments in period t
- $Y_t - T_t$ is disposable income
- $\bar{Y}_t - \bar{T}_t$ is normal disposable income

Propagation of economic fluctuations

Follow the same steps as in previous notes to derive the IS curve:

(divide the national income identity by $\bar{Y}_t$ and subtract 1 from both sides)

We get:

$$\tilde{Y}_t = \frac{1}{1-\xi} [(\alpha_C + \alpha_G + \alpha_I - 1) - \beta(r_t - \bar{r}) - \xi(\tau_t - \bar{\tau})]$$

where $\tau_t = T_t/\bar{Y}_t$ and $\bar{\tau} = \bar{T}_t/\bar{Y}_t$ are the current and normal tax-to-GDP ratios

Propagation of economic fluctuations

This is the same IS curve as before (if we adjust for international trade) with two exceptions:

- We now multiply everything by $\frac{1}{1-\xi}$
 - Note that $0 \leq \xi < 1$
 - so $\frac{1}{1-\xi} > 1$ (if $\xi > 0$)
 - The IS curve is flatter than before
- There is a term relating to taxation
 - If taxes increase, consumption falls and so does output

Multipliers

The marginal propensity to consume

Why do we get the new term $\frac{1}{1-\xi} > 1$ in the IS curve?

- Recall that ξ shows how consumption responds to short-term fluctuations in disposable income
- ξ is often referred to as the **marginal propensity to consume**, MPC, out of income
- Suppose that government spending increases by one unit:
 - Then output Y_t also increases by one unit
 - Higher output means that household income increases – consumption then increases by ξ units
 - This in turn raises income further and results in consumption increasing by an additional ξ^2 units
 - In total, income will increase by $1 + \xi + \xi^2 + \xi^3 + \dots$ units
 - Recall infinite geometric sum (when $0 \leq \xi < 1$):

$$1 + \xi + \xi^2 + \xi^3 + \dots = \frac{1}{1 - \xi}$$

Multipliers

- The positive MPC is an important ingredient in Keynesian business cycle analysis
- Clear that fluctuations can be problematic here, and that policy can be helpful
- Suppose for example that house prices fall:
 - Households feel poorer
 - They want to save more to rebuild wealth
 - What happens to the economy and to household wealth?
- Highlights differences between micro and macro

The slope of the IS curve

Why does the IS curve become flatter when the MPC is higher?

- Recall that the IS curve is: $\tilde{Y}_t = \dots - \frac{\beta}{1-\xi} r_t + \dots$
- In other words, $r_t = \dots - \frac{1-\xi}{\beta} \tilde{Y}_t$
- So the slope of the IS curve is $-\frac{1-\xi}{\beta}$
- Suppose that the central bank cuts the interest rate
 - The MP curve shifts down and output increases since firms invest more
 - If $\xi > 0$, households now raise their consumption
 - ... which further adds to demand in the economy and results in even higher output

AS-AD

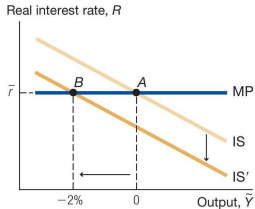
Recall IS-MP-PC

The IS-MP-PC framework helps us understand how:

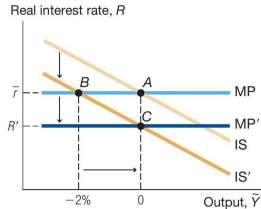
- monetary policy (the MP line) interacts with the economy's demand side (the IS curve) in the very short run, and
- how this in turn leads to movements along the Phillips curve and changes in inflation

See graph on next page

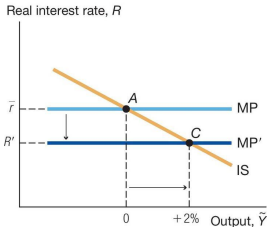
Recall IS-MP-PC



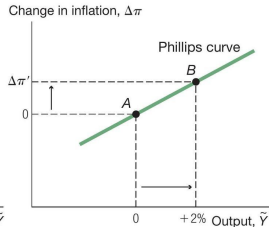
(a) Policymakers believe the slow-down in growth is caused by a negative aggregate demand shock...



(b) so they lower R .



(c) In truth, the IS curve never shifted...



(d) so $\downarrow R \Rightarrow \uparrow \tilde{Y} \Rightarrow \uparrow \pi$.

Figure 12.21 in Jones

- The AS-AD model assumes a typical monetary policy response, a **monetary policy rule**
- Combine the policy rule with the IS curve to derive the AD curve (for "aggregate demand")
- The other part of the model, the AS curve (for "aggregate supply"), is just the Phillips curve with supply shocks

The AD curve

IS curve:

$$\tilde{Y}_t = \alpha - \beta(r_t - \bar{r})$$

Policy rule:

$$r_t - \bar{r} = \mu(\pi_t - \pi^*)$$

The policy rule shows how the central bank typically sets a high interest rate when inflation is above the inflation target (π^*)

μ shows how strongly the central bank responds to inflation deviations from target

Use the policy rule in the IS curve to get the AD curve:

$$\tilde{Y}_t = \alpha - \beta\mu(\pi_t - \pi^*)$$

The AD curve

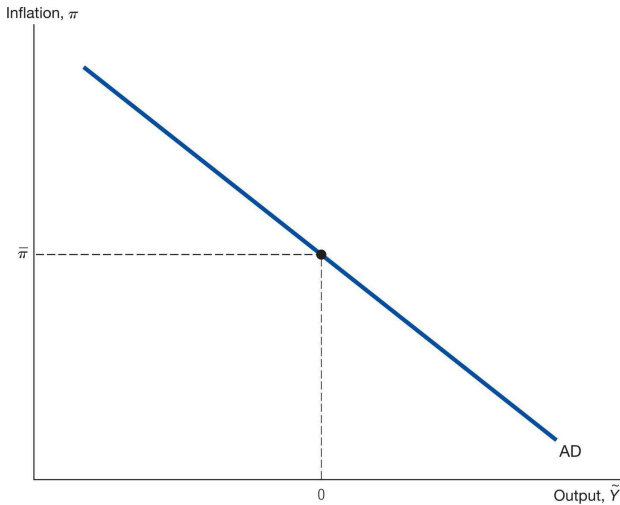


Figure 13.1 in Jones

The AD curve

- Changes in μ : affect slope of AD curve
 - The slope is $-1/(\beta\mu)$
 - ... so the AD curve is flat if μ is large
- Changes in π_t : induce movements along AD curve
- Changes in α : demand shocks that shift the AD curve
 - Higher α (higher "demand") means that the AD curve shifts out

Aggressive monetary policy rule

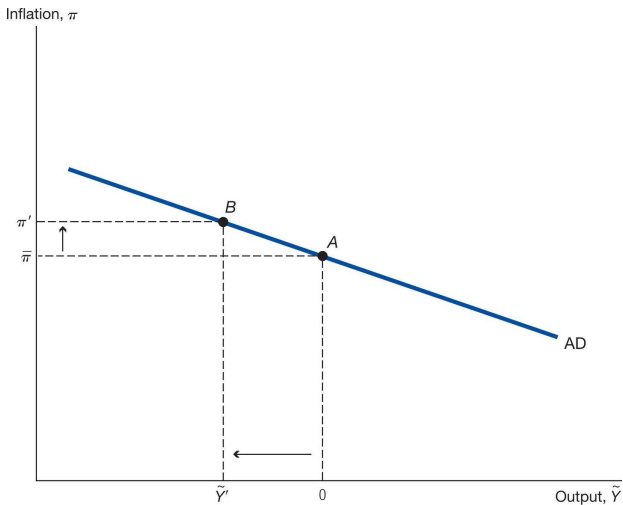


Figure 13.3 in Jones

AS curve

Recall the Phillips curve with supply shocks from the previous lecture notes:

$$\pi_t = \pi_t^e + \nu \tilde{Y}_t + \sigma_t$$

We assumed that inflation expectations are adaptive: $\pi_t^e = \pi_{t-1}$

The Phillips curve with adaptive expectations and supply shocks:

$$\pi_t = \pi_{t-1} + \nu \tilde{Y}_t + \sigma_t$$

This is the AS curve!

AS-AD

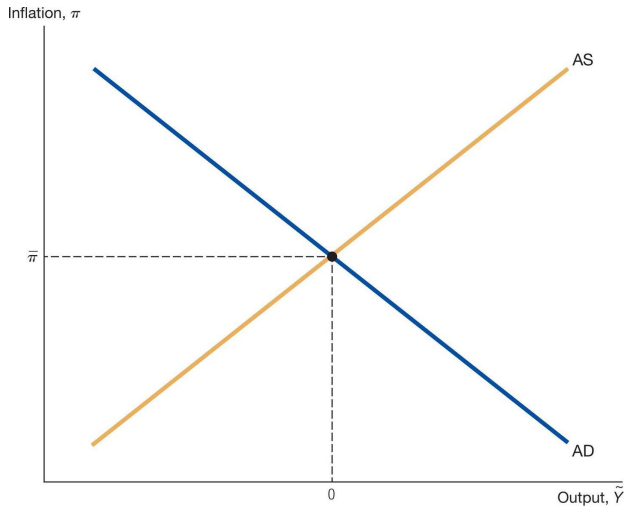


figure 13.5 in Jones

A temporary inflation shock in the AS-AD framework

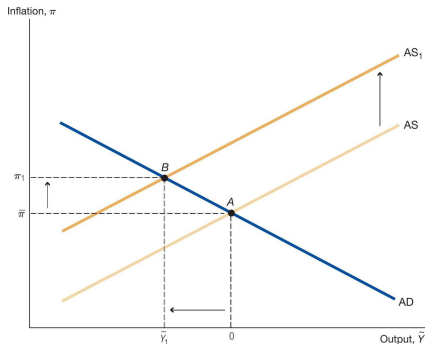


Figure 13.6 in Jones

- The economy was hit by a positive inflation shock, shifting the AS curve up
- $\pi_1 = \pi_0 + \nu \tilde{Y}_1 + \sigma_1$, $\sigma_1 > 0$
- Recall that monetary policy response is built in to the AD curve!
- In B, the central bank has already raised the policy rate and induced a slowdown in economic activity

A temporary inflation shock in the AS-AD framework

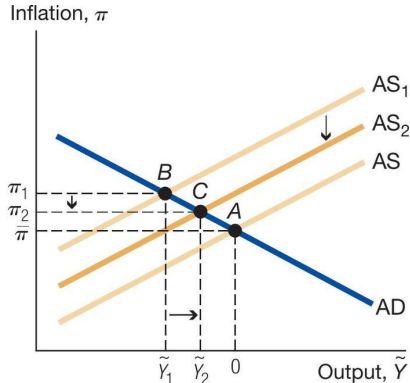
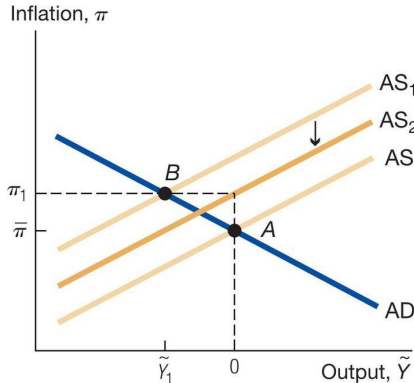


Figure 13.7 in Jones

- The inflation shock was temporary ($\sigma_1 > 0$), but $\sigma_2 = 0$
- But now inflation expectations are high: $\pi_2^e = \pi_1 > \bar{\pi}$
- The AS curve is therefore still above the original AS: $\pi_2 = \pi_1 + \nu \tilde{Y}_2$
- Inflation falls because the output gap is negative, $\tilde{Y}_2 < 0$

Some details on how the AS curve shifts

Consider the AS curves on the previous slide:

- AS_1 shifted up because $\sigma_1 > 0$
 - That curve is σ_1 above the original AS curve
- AS_2 has shifted up relative to the original AS curve because $\pi_2^e = \pi_1 > \bar{\pi}$
 - Note that the output gap $\tilde{Y}_2$ is closed if $\pi_2 = \pi_2^e$
 - AS_2 must therefore lie $\pi_1 - \bar{\pi}$ above the original AS curve
- AS_3 is still above the original AS curve, but it has moved down relative to AS_2
 - It is $\pi_2 - \bar{\pi}$ above the original AS curve

A temporary inflation shock in the AS-AD framework

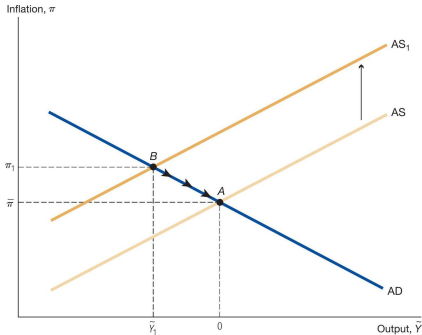


Figure 13.9 in Jones

- As inflation falls, so do inflation expectations and the AS curve shifts down
- As inflation falls, the central bank reduces the policy rate and the economy moves down along the AD curve
- The economy gradually moves closer to the steady state

Disinflation in the AS-AD framework

Suppose that the economy is in an initial equilibrium with high inflation

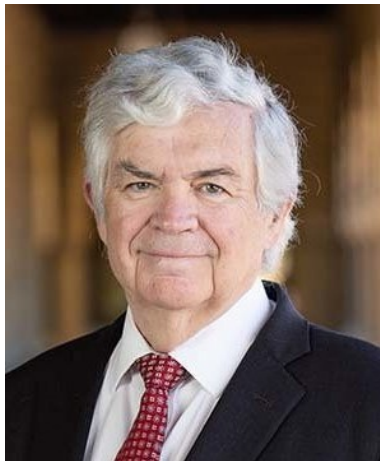
- Think for example of the U.S. or Sweden in the 1970s
- ... and think of this as being an equilibrium with a high π^*

Suppose now that we want to implement a new, lower, inflation target

- What does the AS-AD model say about this?
- What if inflation expectations were not adaptive?

The Taylor rule

How do central banks set their policy rate?



John Taylor, 1946-

- Taylor noted in 1993 that economic outcomes in the U.S. had been good when the Fed responded strongly to inflation (and to the output gap)
- $i_t = \pi_t + \bar{r}_t + \mu(\pi_t - \pi^*) + \omega \tilde{Y}_t$

The Taylor rule

$$i_t = \pi_t + \bar{r}_t + \mu(\pi_t - \pi^*) + \omega \tilde{Y}_t$$

- According to Taylor (1993), a good rule would be $\mu = 0.5$ and $\omega = 0.5$
 - Note that $\mu = 0.5$ means that i_t is raised by 1.5 percentage points if π_t increases by one percentage point
- The idea that monetary policy must respond strongly to inflation is widely accepted
- Debated how to respond to the output gap
 - The output gap is difficult to measure
 - Some research indicate that strong reactions to such a vague variable can do more harm than good

The original Taylor rule for 1987-1992

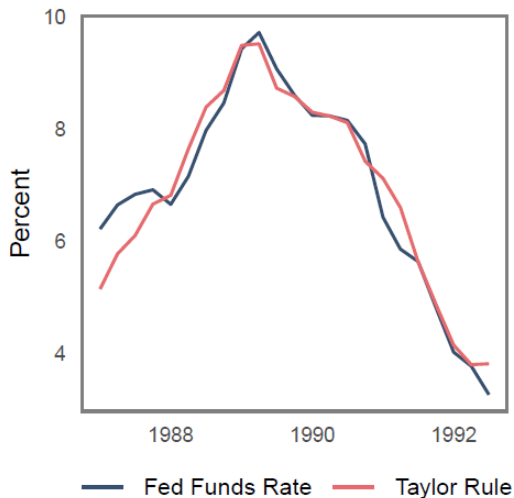


Figure 2: Fit of Original Taylor Rule, 1987-1992

The Taylor principle

Recall the Fisher equation:

$$r_t = i_t - \pi_t^e$$

- Suppose that $\pi_t^e = \pi_t$
- Then if π_t increases, i_t must increase *more* for the real interest rate to increase
- In other words, $\mu > 0$ is necessary for the real interest rate to respond in the right direction
- This – that we must have $\mu > 0$ – is known as the **Taylor principle**

The Taylor rule

- Of course, central banks do *not* follow simple rules of thumb
- But much of how central banks think can be illustrated with a policy rule like the Taylor rule
- We will dig a bit further into what central banks actually do in the next set of lecture notes

The original Taylor rule for 1965-2025

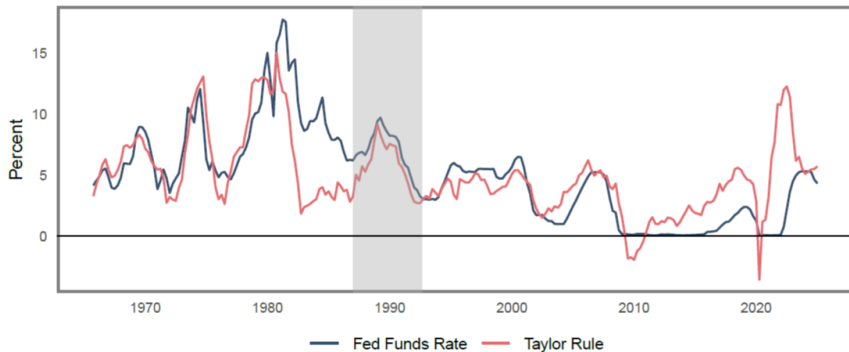


Figure 3: Original Taylor Rule with Real-Time Data

Note: The (light) red line shows the policy rate implied by the original Taylor rule—equation (1). We use real-time data for the GDP deflator and the Greenbook's measure of the output gap (CBO output gap for last few years). The sample period is 1965Q4-2025Q1. The shaded area shows the sample period considered in Taylor (1993).

Figure 3 in Nakamura et al. (2025)

Rules vs discretion and the time inconsistency problem of monetary policy

Rules vs discretion

- Should monetary policy follow strict "rules" similar to the Taylor rule?
- Or should monetary policy be discretionary: "re-optimize" in each period taking all available information into account
- Most economists agree that following strict rules would be a bad idea
- But the time inconsistency problem of monetary policy illustrates potential problems with discretionary policy

The time inconsistency problem of monetary policy

- Kydland and Prescott (1977): policymakers' inability to commit to low inflation may give rise to excessive inflation¹
- In the absence of a binding commitment to low inflation, inflation may increase without any corresponding increase in output

¹"Rules rather than discretion: The inconsistency of optimal plans", Journal of Political Economy. See the [motivation for 2004 Nobel prize](#).

The time inconsistency problem of monetary policy

Consider a simple one-period model characterized by this Phillips curve:

$$\pi = \pi^e + \nu \tilde{Y}, \quad (1)$$

where π and $\tilde{Y}$ denote inflation and the output gap, and π^e is expected inflation

To simplify as much as possible, suppose that the policymaker can *choose* the inflation rate

Moreover, suppose that the inflation rate can be chosen after inflation expectations have been formed (e.g., after prices are set and wage contracts negotiated)

The time inconsistency problem of monetary policy

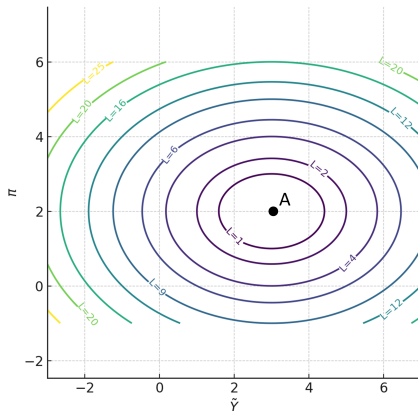
- Suppose that the policymaker wants inflation to be $\pi = \pi^*$ and the output gap to be $\tilde{Y} = Y^* > 0$

Suppose for example a loss function, $L = (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2$

- The policymaker may then try to "surprise" the market by letting inflation be higher than expected
- But the market will anticipate this (or at least eventually learn about it) and therefore expect high inflation
- The equilibrium outcome will be that $\tilde{Y} = 0$ but $\pi > \pi^*$
- There will be an *inflation bias* in this economy

The time inconsistency problem of monetary policy

- The loss function can be represented by indifference curves
- Combinations $(\pi, \tilde{Y})$ on the same curve result in the same loss
- The policymaker prefers point A
- Curves further away from A are worse

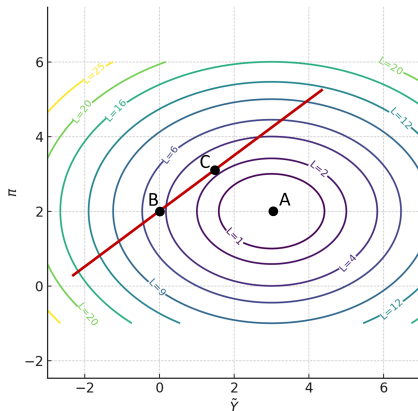


$$L = (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2$$

$$\pi^* = 2, \lambda = 0.5 \text{ and } Y^* = 3$$

The time inconsistency problem of monetary policy

- Suppose that $\pi^e = 2$
- The Phillips curve is then the red line
- The policymaker prefers point C over point B
- People anticipate this, so $\pi^e = 2$ is not possible

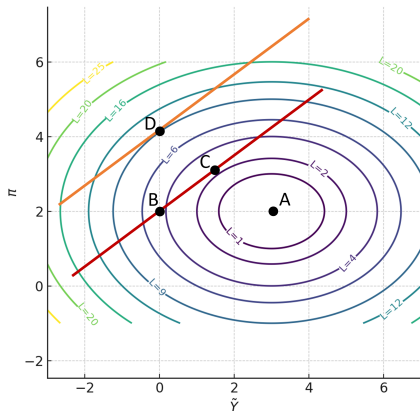


$$L = (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2$$

$$\pi^* = 2, \lambda = 0.5 \text{ and } Y^* = 3$$

The time inconsistency problem of monetary policy

- Consider point D where $\pi^e \approx 4$
- The Phillips curve is then the orange line
- D is the best point on that Phillips curve
- This is the equilibrium
 - Here $\pi > \pi^*$
 - ... but output has not increased ($\tilde{Y} = 0$)
- The outcome would be better if the policymaker did not care about $\tilde{Y}$



$$L = (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2$$

$$\pi^* = 2, \lambda = 0.5 \text{ and } Y^* = 3$$

The time inconsistency problem: analytical solution

The policymaker takes π^e as given and minimizes the loss function,

$$\min_{\pi} (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2$$

subject to the Phillips curve

$$\pi = \pi^e + \nu \tilde{Y}$$

The time inconsistency problem: analytical solution

The Lagrangian is:

$$\mathcal{L} = (\pi - \pi^*)^2 + \lambda(\tilde{Y} - Y^*)^2 + \mu(\pi^e + \nu\tilde{Y} - \pi)$$

where μ is the lagrange multiplier

The first order conditions with respect to π and $\tilde{Y}$ are then:

$$2(\pi - \pi^*) - \mu = 0$$

and

$$2\lambda(\tilde{Y} - Y^*) + \mu\nu = 0$$

The time inconsistency problem: analytical solution

Use the first order conditions together with the Phillips curve. We see that the policymaker should "set inflation" so that

$$\left(1 + \frac{\lambda}{\nu^2}\right) \pi = \pi^* + \frac{\lambda}{\nu^2} \pi^e + \frac{\lambda}{\nu} Y^*$$

Since all this is anticipated when households and firms form inflation expectations, it must be the case that $\pi^e = \pi$

Using this observation in the equation above, we see that equilibrium inflation will be

$$\pi = \pi^* + \frac{\lambda}{\nu} Y^*$$

If $Y^* > 0$ and $\lambda > 0$, inflation will be higher than the target!

Solving the time-inconsistency problem

- One suggested solution: Delegate monetary policy to an independent central banker that has other preferences than politicians/society
 - a "conservative" central banker that cares less about unemployment
- A better idea: Set up institutions to eliminate root of the problem:
 - Delegate monetary policy to independent central bank
 - Formulate (politicians can do this) clear objectives, the target
 - Committee with overlapping terms (avoid final-period problem)
 - Communicate transparently: the target, tools used, analysis/forecasts/motivation
 - All this allows the central bank to build credibility, accumulate a reputation, to not exploit the Phillips curve